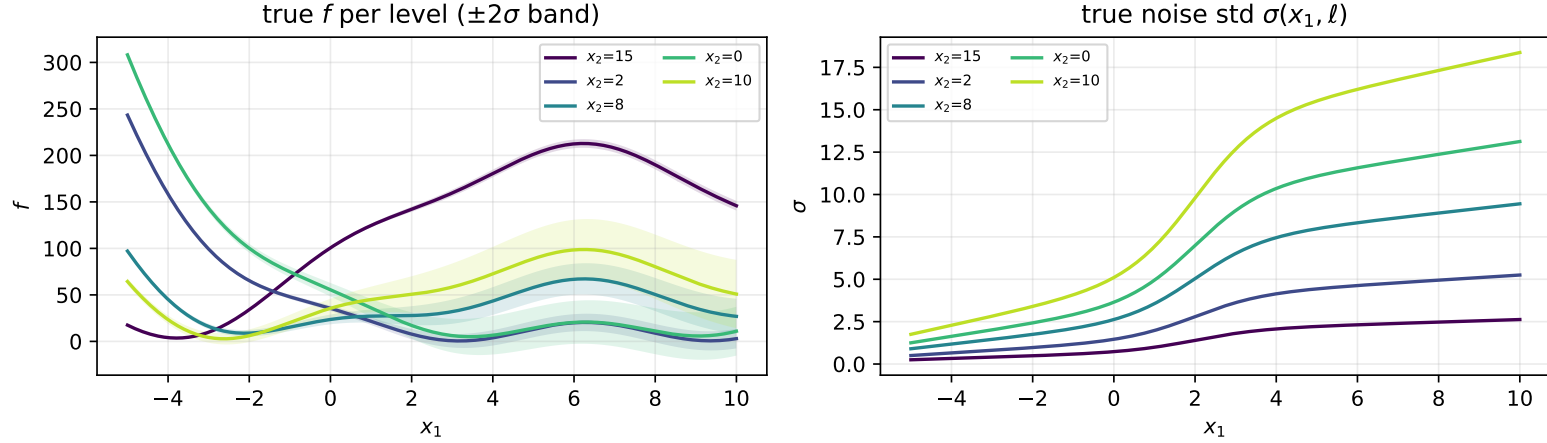


Test problems

Branin (heteroscedastic, 5 levels)

$$f(x_1, \ell) = \left(x_2(\ell) - \frac{5.1}{4\pi^2} x_1^2 + \frac{5}{\pi} x_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos x_1 + 10, \quad x_1 \in [-5, 10], \quad x_2(\ell) \in \{15, 2, 8, 0, 10\}$$

$$\sigma(x_1, \ell) = \left[0.5 + 0.15(x_1 + 5) + \frac{2.5}{1 + e^{-1.2(x_1 - 2)}} \right] m(\ell), \quad m(\ell) \in \{0.5, 1.0, 1.8, 2.5, 3.5\}$$

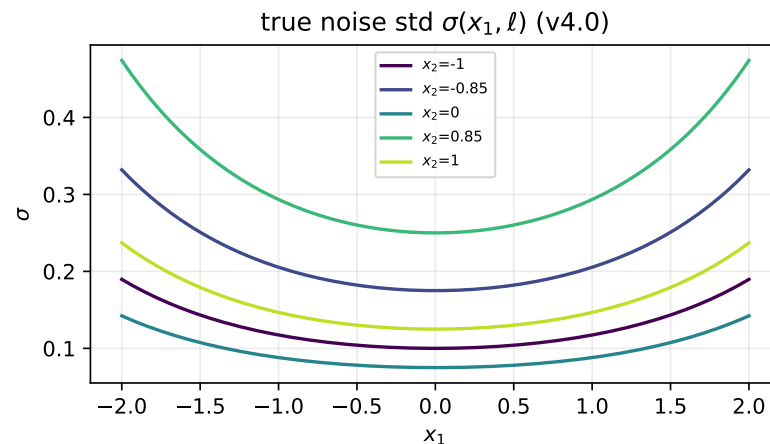
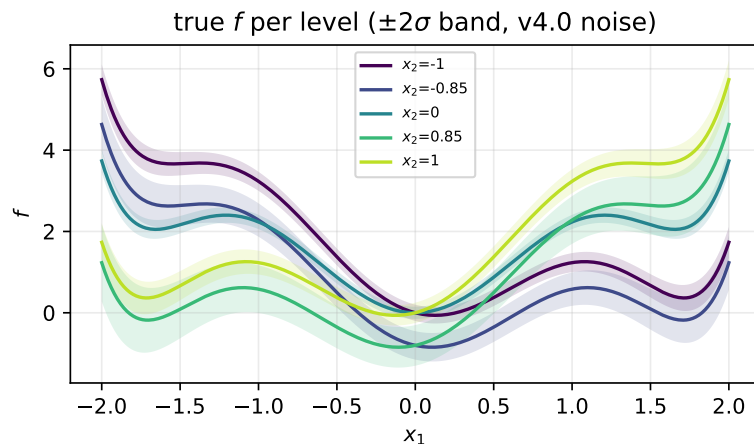


Six-hump camel (5 levels: two pairs + bridge)

$$f(x_1, \ell) = \left(4 - 2.1x_1^2 + \frac{x_1^4}{3} \right) x_1^2 + x_1 x_2(\ell) + (-4 + 4x_2(\ell)^2) x_2(\ell)^2, \quad x_1 \in [-2, 2], \quad x_2(\ell) \in \{-1.0, -0.85, 0.0, 0.85, 1.0\}$$

$$\sigma(x_1, \ell) = 0.05 e^{(0.4x_1)^2} m(\ell), \quad m(\ell) \in \{2.0, 3.5, 1.5, 5.0, 2.5\}$$

(noise as used in the 30-seed replication tables below)



Griewank 6-D (level-shifted pairs)

$$f(\mathbf{x}_q, \ell) = \sum_{i=1}^6 \frac{x s_i^2}{4000} - \prod_{i=1}^6 \cos\left(\frac{x s_i}{\sqrt{i}}\right) + 1 + b(\ell), \quad x s = [\mathbf{x}_q, v(\ell)], \quad \mathbf{x}_q \in [-3, 3]^5$$

$$v(\ell) \in \{0.5, 1.3, 2.7, 3.1\}, \quad b(\ell) \in \{0, 0.15, 0.6, 0.75\}, \quad \sigma(\mathbf{x}_q, \ell) = 0.01 \cdot 10^{(x_1+3)/6} m(\ell), \quad m(\ell) \in \{1.5, 1.0, 3.0, 2.0\}$$

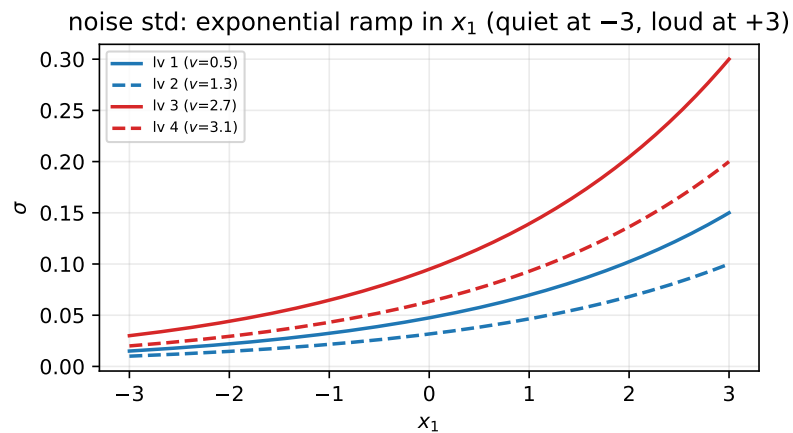
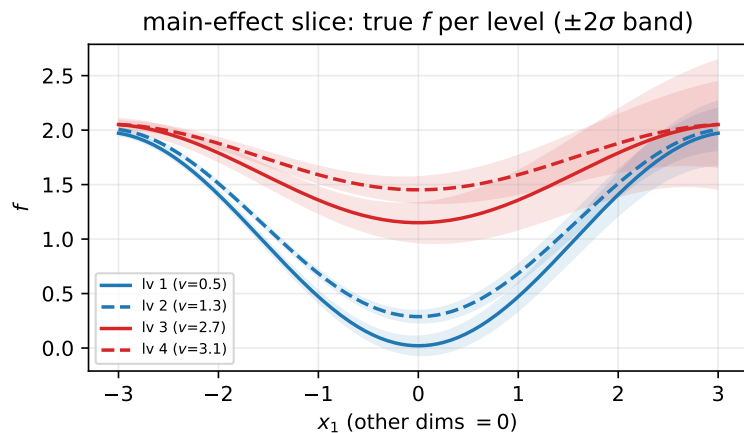


Table 1: 30-seed replication (fixed design, fresh noise replicates per seed): mean $\pm$ std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Branin	25	Separate GPs	10.91 $\pm$ 0.466	0.434 $\pm$ 0.022	113.7 $\pm$ 11.60	0.775 $\pm$ 0.028
		Categorical GP	4.529 $\pm$ 0.241	0.159 $\pm$ 0.011	38.87 $\pm$ 2.018	0.946 $\pm$ 0.010
		Standard LVGP	4.419 $\pm$ 1.063	0.212 $\pm$ 0.051	74.80 $\pm$ 30.69	0.442 $\pm$ 0.143
		H-LVGP	2.520 $\pm$ 0.723	0.096 $\pm$ 0.037	37.28 $\pm$ 20.11	0.996 $\pm$ 0.016
	40	Separate GPs	4.454 $\pm$ 0.361	0.208 $\pm$ 0.017	42.88 $\pm$ 1.784	0.931 $\pm$ 0.011
		Categorical GP	1.585 $\pm$ 0.268	0.054 $\pm$ 0.012	14.86 $\pm$ 0.896	0.996 $\pm$ 0.010
		Standard LVGP	3.109 $\pm$ 0.705	0.122 $\pm$ 0.030	62.82 $\pm$ 20.09	0.410 $\pm$ 0.145
		H-LVGP	1.722 $\pm$ 0.425	0.064 $\pm$ 0.019	35.61 $\pm$ 10.83	0.998 $\pm$ 0.008
Six-hump camel	25	Separate GPs	0.231 $\pm$ 0.009	0.319 $\pm$ 0.007	2.072 $\pm$ 0.061	0.952 $\pm$ 0.008
		Categorical GP	0.168 $\pm$ 0.009	0.262 $\pm$ 0.010	1.778 $\pm$ 0.136	0.923 $\pm$ 0.012
		Standard LVGP	0.152 $\pm$ 0.018	0.242 $\pm$ 0.029	3.405 $\pm$ 0.521	0.417 $\pm$ 0.097
		H-LVGP	0.178 $\pm$ 0.012	0.287 $\pm$ 0.016	2.153 $\pm$ 0.231	0.905 $\pm$ 0.017
	40	Separate GPs	0.196 $\pm$ 0.008	0.295 $\pm$ 0.006	2.313 $\pm$ 0.111	0.888 $\pm$ 0.015
		Categorical GP	0.114 $\pm$ 0.010	0.160 $\pm$ 0.015	1.127 $\pm$ 0.164	0.918 $\pm$ 0.013
		Standard LVGP	0.145 $\pm$ 0.034	0.212 $\pm$ 0.066	3.240 $\pm$ 1.092	0.324 $\pm$ 0.106
		H-LVGP	0.143 $\pm$ 0.024	0.226 $\pm$ 0.042	1.147 $\pm$ 0.318	0.953 $\pm$ 0.029
Griewank 6-D	48	Separate GPs	0.076 $\pm$ 0.003	1.086 $\pm$ 0.036	0.833 $\pm$ 0.034	0.861 $\pm$ 0.014
		Categorical GP	0.066 $\pm$ 0.003	0.939 $\pm$ 0.048	0.695 $\pm$ 0.027	0.893 $\pm$ 0.015
		Standard LVGP	0.104 $\pm$ 0.009	1.376 $\pm$ 0.134	1.385 $\pm$ 0.193	0.716 $\pm$ 0.052
		H-LVGP	0.097 $\pm$ 0.007	1.242 $\pm$ 0.169	0.939 $\pm$ 0.070	0.845 $\pm$ 0.021
	64	Separate GPs	0.066 $\pm$ 0.002	0.973 $\pm$ 0.025	0.773 $\pm$ 0.031	0.882 $\pm$ 0.010
		Categorical GP	0.064 $\pm$ 0.003	0.896 $\pm$ 0.041	0.748 $\pm$ 0.022	0.912 $\pm$ 0.008
		Standard LVGP	0.086 $\pm$ 0.012	1.210 $\pm$ 0.168	1.203 $\pm$ 0.273	0.736 $\pm$ 0.043
		H-LVGP	0.084 $\pm$ 0.026	1.141 $\pm$ 0.321	0.916 $\pm$ 0.226	0.876 $\pm$ 0.029
	80	Separate GPs	0.071 $\pm$ 0.002	1.016 $\pm$ 0.040	0.730 $\pm$ 0.028	0.871 $\pm$ 0.010
		Categorical GP	0.061 $\pm$ 0.003	0.986 $\pm$ 0.046	0.723 $\pm$ 0.024	0.887 $\pm$ 0.009
		Standard LVGP	0.100 $\pm$ 0.010	1.359 $\pm$ 0.145	1.084 $\pm$ 0.228	0.820 $\pm$ 0.047
		H-LVGP	0.082 $\pm$ 0.015	1.133 $\pm$ 0.277	0.950 $\pm$ 0.089	0.836 $\pm$ 0.047
	120	Separate GPs	0.064 $\pm$ 0.003	0.903 $\pm$ 0.036	0.690 $\pm$ 0.063	0.880 $\pm$ 0.017
		Categorical GP	0.050 $\pm$ 0.001	0.749 $\pm$ 0.020	0.539 $\pm$ 0.015	0.922 $\pm$ 0.008
		Standard LVGP	0.101 $\pm$ 0.015	1.382 $\pm$ 0.186	1.283 $\pm$ 0.269	0.773 $\pm$ 0.048
		H-LVGP	0.111 $\pm$ 0.033	1.564 $\pm$ 0.605	1.101 $\pm$ 0.182	0.856 $\pm$ 0.031
	160	Separate GPs	0.056 $\pm$ 0.002	0.851 $\pm$ 0.040	0.683 $\pm$ 0.052	0.836 $\pm$ 0.018
		Categorical GP	0.047 $\pm$ 0.003	0.704 $\pm$ 0.053	0.598 $\pm$ 0.024	0.886 $\pm$ 0.011
		Standard LVGP	0.075 $\pm$ 0.008	1.048 $\pm$ 0.107	0.948 $\pm$ 0.391	0.811 $\pm$ 0.100
		H-LVGP	0.137 $\pm$ 0.027	2.091 $\pm$ 0.433	1.047 $\pm$ 0.132	0.934 $\pm$ 0.020

Table 2: 30-seed replication (fixed design, fresh noise replicates per seed): mean $\pm$ std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95). Without the Categorical GP (3-model comparison; winners recomputed).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Branin	25	Separate GPs	10.91 $\pm$ 0.466	0.434 $\pm$ 0.022	113.7 $\pm$ 11.60	0.775 $\pm$ 0.028
		Standard LVGP	4.419 $\pm$ 1.063	0.212 $\pm$ 0.051	74.80 $\pm$ 30.69	0.442 $\pm$ 0.143
		H-LVGP	2.520 $\pm$ 0.723	0.096 $\pm$ 0.037	37.28 $\pm$ 20.11	0.996 $\pm$ 0.016
	40	Separate GPs	4.454 $\pm$ 0.361	0.208 $\pm$ 0.017	42.88 $\pm$ 1.784	0.931 $\pm$ 0.011
		Standard LVGP	3.109 $\pm$ 0.705	0.122 $\pm$ 0.030	62.82 $\pm$ 20.09	0.410 $\pm$ 0.145
		H-LVGP	1.722 $\pm$ 0.425	0.064 $\pm$ 0.019	35.61 $\pm$ 10.83	0.998 $\pm$ 0.008
Six-hump camel	25	Separate GPs	0.231 $\pm$ 0.009	0.319 $\pm$ 0.007	2.072 $\pm$ 0.061	0.952 $\pm$ 0.008
		Standard LVGP	0.152 $\pm$ 0.018	0.242 $\pm$ 0.029	3.405 $\pm$ 0.521	0.417 $\pm$ 0.097
		H-LVGP	0.178 $\pm$ 0.012	0.287 $\pm$ 0.016	2.153 $\pm$ 0.231	0.905 $\pm$ 0.017
	40	Separate GPs	0.196 $\pm$ 0.008	0.295 $\pm$ 0.006	2.313 $\pm$ 0.111	0.888 $\pm$ 0.015
		Standard LVGP	0.145 $\pm$ 0.034	0.212 $\pm$ 0.066	3.240 $\pm$ 1.092	0.324 $\pm$ 0.106
		H-LVGP	0.143 $\pm$ 0.024	0.226 $\pm$ 0.042	1.147 $\pm$ 0.318	0.953 $\pm$ 0.029
Griewank 6-D	48	Separate GPs	0.076 $\pm$ 0.003	1.086 $\pm$ 0.036	0.833 $\pm$ 0.034	0.861 $\pm$ 0.014
		Standard LVGP	0.104 $\pm$ 0.009	1.376 $\pm$ 0.134	1.385 $\pm$ 0.193	0.716 $\pm$ 0.052
		H-LVGP	0.097 $\pm$ 0.007	1.242 $\pm$ 0.169	0.939 $\pm$ 0.070	0.845 $\pm$ 0.021
	64	Separate GPs	0.066 $\pm$ 0.002	0.973 $\pm$ 0.025	0.773 $\pm$ 0.031	0.882 $\pm$ 0.010
		Standard LVGP	0.086 $\pm$ 0.012	1.210 $\pm$ 0.168	1.203 $\pm$ 0.273	0.736 $\pm$ 0.043
		H-LVGP	0.084 $\pm$ 0.026	1.141 $\pm$ 0.321	0.916 $\pm$ 0.226	0.876 $\pm$ 0.029
	80	Separate GPs	0.071 $\pm$ 0.002	1.016 $\pm$ 0.040	0.730 $\pm$ 0.028	0.871 $\pm$ 0.010
		Standard LVGP	0.100 $\pm$ 0.010	1.359 $\pm$ 0.145	1.084 $\pm$ 0.228	0.820 $\pm$ 0.047
		H-LVGP	0.082 $\pm$ 0.015	1.133 $\pm$ 0.277	0.950 $\pm$ 0.089	0.836 $\pm$ 0.047
	120	Separate GPs	0.064 $\pm$ 0.003	0.903 $\pm$ 0.036	0.690 $\pm$ 0.063	0.880 $\pm$ 0.017
		Standard LVGP	0.101 $\pm$ 0.015	1.382 $\pm$ 0.186	1.283 $\pm$ 0.269	0.773 $\pm$ 0.048
		H-LVGP	0.111 $\pm$ 0.033	1.564 $\pm$ 0.605	1.101 $\pm$ 0.182	0.856 $\pm$ 0.031
	160	Separate GPs	0.056 $\pm$ 0.002	0.851 $\pm$ 0.040	0.683 $\pm$ 0.052	0.836 $\pm$ 0.018
		Standard LVGP	0.075 $\pm$ 0.008	1.048 $\pm$ 0.107	0.948 $\pm$ 0.391	0.811 $\pm$ 0.100
		H-LVGP	0.137 $\pm$ 0.027	2.091 $\pm$ 0.433	1.047 $\pm$ 0.132	0.934 $\pm$ 0.020